PANDIT DEENDAYAL ENERGY UNIVERSITY • SCHOOL OF TECHNOLOGY

PROJECT TRASURE

B2B Circular Economy Auction Engine & By-Product Trade Platform

(Web Platform Architecture & Engineering Implementation)

Final Software Engineering Project Report

Course: Software Engineering Laboratory & Theory

SUBMITTED SINGLE-HANDEDLY BY

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Table of Contents

Section	Page / Reference
1. Roles and Contributions	**2**
2. SDLC Methodology	**3**
3. Project Abstract	**5**
4. Questionnaire / Requirement Clarification	**6**
5. Functional Requirements	**7**
6. Non-Functional Requirements and Main Modules	**8**
7. UML and Architecture Diagrams	**9–14**
7.1 Class Diagram	**9**
7.2 Sequence Diagram	**10**
7.3 Use Case Diagram	**11**
7.4 Activity Diagram (Part 1 & Part 2)	**12–13**
7.5 Component Diagram	**14**
7.6 Deployment Diagram	**14**
8. Test Cases and Results	**15–18**
8.1 Niv Sorathiya — Unit Tests	**15–16**
8.2 System Tests	**17**
8.3 Defects and Corrections	**18**
9. Testing Evidence	**19–21**
10. Conclusion and Repository Link	**22**
11. Student / Solo Reflection	**23**

1. Roles and Contributions

Project TRASURE was architected, engineered, styled, tested, and documented **single-handedly by Niv Sorathiya**. Every phase of the engineering lifecycle—from industrial circular economy domain research, mathematical modeling of delivered landed tariffs, interactive GIS road routing, reverse auction state modeling, test automation, and final report compilation—was executed independently.

Student Name	Roll Number	Degree & Branch	Key Responsibilities & Project Contributions
Niv Sorathiya	**24BCB032**	B.Tech – CSBS PDEU (2026)	**Full Lifecycle Ownership:** • Authored requirements specification & stakeholder questionnaires. • Designed non-functional and regulatory architecture (SPCB compliance). • Modeled all 6 student-level UML and architecture diagrams. • Implemented Delivered Landed Cost formula, GIS road transit router, reverse auction engine with anti-sniping, and auto-settlement invoicing. • Performed unit testing, system-level verification, defect correction, and user manual compilation.

Solo Scope Governance: To ensure high maintainability as an independent developer, strict modular decoupling was enforced from inception. The web application was divided into discrete functional controllers (`LandedCostEngine`, `FreightGISRouter`, `ReverseAuctionDesk`, `ComplianceManager`, `InvoiceBillingModule`, and `StorageManager`), allowing each workflow to be developed and verified iteratively without introducing regression bugs.

2. SDLC Methodology: Iterative and Incremental Approach

Project TRASURE was engineered using an **Iterative and Incremental Software Development Life Cycle (SDLC)** approach. Given the mathematical interdependency between delivered landed costs, real-time spatial road routing APIs, asynchronous ticker timers, and statutory compliance documentation, constructing the platform in discrete, self-contained iterations allowed early defect discovery and progressive refinement.

Requirements Discovery → Architectural UML Modeling → Iterative Cycles (Landed Cost → GIS Routing → Live Auction → Compliance → Billing) → Integration → Verification → Release

Development Stages

1. **Requirement Identification:** Analyzed secondary industrial by-product streams (Fly Ash, GGBS Slag, Spent Acid, Phosphogypsum). Identified core operational pain points: high bulk freight tariffs masking ex-factory pricing, lack of dynamic reverse auction discovery, and mandatory SPCB Form 10 hazardous waste manifests.
2. **Planning and Architectural Design:** Established clean object boundaries. Designed student-level UML models adhering to standard object-oriented design patterns to balance academic rigor with clean browser execution.
3. **Incremental Implementation:**
    - *Iteration 1:* Catalog filtering matrix and Delivered Landed Cost calculation engine.
    - *Iteration 2:* Leaflet GIS mapping with Esri World Gray Canvas tiles and OSRM turn-by-turn road network routing.
    - *Iteration 3:* Live B2B reverse auction desk featuring real-time tickers, bid undercutting guards, and 60-second anti-sniping overtime extensions.
    - *Iteration 4:* SPCB environmental compliance module for Form 10 hazardous waste movement tracking.
    - *Iteration 5:* Automated commercial tax invoicing with 18% dual GST (9% CGST + 9% SGST).
4. **Integration & Client Persistence:** Connected all controllers via a unified sandboxed browser storage layer (`localStorage`) with automated seed data recovery.
5. **Testing & Defect Correction:** Conducted 10 unit tests and 6 multi-system workflows. Logged defects (such as map tile authorization blockades, UI currency unit clipping, and auction race conditions) and verified corrective actions.
6. **Final Documentation:** Compiled academic reports, vector UML architecture diagrams, and comprehensive user manuals.

Rationale for Selecting This Approach

A traditional linear Waterfall model would have introduced significant risk due to the unknown latency and rate limits of external GIS mapping APIs and the asynchronous state demands of live auction timers. The iterative approach enabled isolated verification of individual components—such as testing the landed cost formula and polyline highway rendering—prior to coupling them with high-concurrency auction bidding.

3. Project Abstract

Project TRASURE is an enterprise B2B circular economy web platform engineered to optimize the discovery, freight transit, price discovery, and regulatory compliance of industrial bulk by-products. Heavy process industries—such as thermal power plants, integrated steel mills, chemical manufacturing complexes, and fertilizer refineries—produce millions of metric tonnes (MT) of secondary materials annually (including Fly Ash, Ground Granulated Blast-Furnace Slag [GGBS], Spent Sulfuric Acid, and Phosphogypsum).

Historically, trading these bulk by-products has been severely impaired by three structural bottlenecks:

- **The Delivered Cost Blindspot:** Ex-factory base quotations are deceptive. Because bulk by-products carry low value-to-weight ratios, long-haul road transport across several hundred kilometers frequently increases the final landed price by 200% to 300%.
- **Opaque Price Discovery:** Static offline negotiations prevent buyers from capitalizing on dynamic regional surplus or competitive supplier discounts.
- **Stringent Regulatory Compliance:** Hazardous and heavy industrial materials require mandatory transit tracking under Central and State Pollution Control Board (CPCB/SPCB) Hazardous and Other Wastes Rules.

TRASURE solves these challenges through an integrated single-page web application featuring:

- **Delivered Landed Cost Engine:** Simultaneously evaluates base price, real road distance, freight haulage rates (₹/km/MT), and loading fees to rank suppliers by actual gate cost.
- **GIS Road Transit Routing:** Leverages high-contrast Esri cartographic tiles and the OSRM routing engine to plot turn-by-turn highway transit corridors across Indian road networks.
- **Live B2B Reverse Auction Exchange:** An automated procurement floor where pre-qualified industrial sellers compete to fulfill buyer tenders, protected by automated anti-sniping timer extensions.
- **SPCB Regulatory Compliance Desk:** Automatically produces standardized Form 10 Hazardous Waste Movement Manifests with verified HAZCHEM codes and transporter details.
- **Automated B2B Tax Invoicing:** Computes statutory 18% GST (9% CGST + 9% SGST) and renders printable commercial invoices.

Engineered purely in standards-compliant HTML5, responsive CSS3, and modern Vanilla ES6+ JavaScript, TRASURE delivers rapid execution with zero external server runtime dependencies.

4. Questionnaire / Requirement Clarification

The following structured questionnaire documents the core functional and architectural decisions established during the requirements engineering phase of Project TRASURE.

#	Requirement Question	Architectural Decision & Technical Solution
1	**What is the primary purpose of the platform?**	To provide an end-to-end B2B circular economy web portal where industrial procurement teams discover, bid upon, and legally transport bulk secondary by-products based on true delivered landed cost rather than misleading ex-factory rates.
2	**How does TRASURE eliminate the landed cost blindspot?**	By implementing an automated Delivered Landed Cost calculation engine: `Delivered Cost = Ex-Plant Price + (Road Distance in km × Freight Tariff per MT/km) + Handling Fee` Suppliers are automatically re-ranked from lowest to highest landed cost at the buyer's gate.
3	**How are logistics transit routes determined and visualized?**	The platform pairs Leaflet GIS with the Open Source Routing Machine (OSRM) highway network API. Selecting any origin factory queries the real-world road network to draw an accurate turn-by-turn highway corridor with true distance and transit hours.
4	**What mechanism prevents unfair auction sniping?**	An automated **Anti-Sniping Overtime Extension rule**. If a vendor submits a valid lower counter-bid with less than 60 seconds remaining on the clock, the auction countdown timer automatically resets by adding 60 seconds.
5	**What validation constraints govern incoming auction bids?**	Bids must satisfy three atomic guards: (1) strictly lower than the current leading bid, (2) decrement meets or exceeds the required minimum step (e.g., ₹50/MT), and (3) strictly positive value (`Bid > 0`).
6	**How is environmental compliance guaranteed for hazardous by-products?**	A dedicated SPCB Compliance Engine verifies State Pollution Control Board Consent to Operate (CTO) credentials, assigns official CPCB HAZCHEM schedule codes, and compiles standardized Form 10 manifests.
7	**How are financial settlements and tax invoices processed?**	Upon auction closure or spot order, the platform computes statutory Indian B2B taxes—splitting 18% GST into 9% CGST and 9% SGST—and generates an itemized, printable commercial tax invoice.
8	**What data persistence architecture is utilized?**	The web client operates 100% client-side using browser `localStorage` with automated fallback seed data mounting, ensuring zero server downtime during demonstrations and offline testing.

5. Functional Requirements

1. Material Sourcing Catalog & Landed Cost Discovery

- **FR-1.1:** System shall present industrial by-product listings categorized into Fly Ash, GGBS Slag, Spent Acid, and Phosphogypsum.
- **FR-1.2:** System shall provide multi-parameter filtering by material category, seller identity, purity/grade, and geographic state.
- **FR-1.3:** System shall calculate and display the total delivered landed cost per metric tonne dynamically based on the buyer's configured receiving terminal.
- **FR-1.4:** System shall rank matching suppliers from lowest delivered landed cost to highest, highlighting the optimal vendor with an "Optimal Landed Cost" badge.

2. GIS Logistics & Highway Route Plotting

- **FR-2.1:** System shall render an interactive Leaflet GIS map utilizing high-contrast Esri World Gray Canvas vector tiles.
- **FR-2.2:** System shall query the OSRM highway routing engine to calculate exact road driving distance (km) and estimated transit hours.
- **FR-2.3:** System shall plot turn-by-turn road transit corridors with smooth directional polylines on the map viewport.
- **FR-2.4:** System shall provide a plant switcher that re-centers the map view, updates the road trajectory, and recalibrates logistics transit statistics immediately.

3. Live B2B Reverse Auction Exchange

- **FR-3.1:** System shall allow procurement officers to initialize live reverse auction sessions for bulk tenders with pre-configured target lots.
- **FR-3.2:** System shall display live ticker countdown clocks, current leading (lowest) bid, baseline ceiling price, and supplier leaderboard.
- **FR-3.3:** System shall validate counter-bids, ensuring they undercut the leading bid by at least the minimum required decremental step.
- **FR-3.4:** System shall enforce Anti-Sniping Protection by adding 60 seconds to the timer whenever a valid bid is placed with less than 60 seconds remaining.
- **FR-3.5:** System shall provide automated competitor bots simulating market pricing behavior at dynamic intervals.

4. SPCB Regulatory Manifest & Compliance Engine

- **FR-4.1:** System shall categorize materials by environmental risk profile (Non-Hazardous By-Product vs. Schedule II/III Regulated Waste).
- **FR-4.2:** System shall verify SPCB Consent to Operate (CTO) and authorization credentials for participating industrial plants.
- **FR-4.3:** System shall generate downloadable, formatted SPCB Form 10 Hazardous Waste Movement Manifests detailing vehicle registration numbers, driver details, and emergency response instructions.

5. Automated Financial Invoicing & Trade Settlement

- **FR-5.1:** System shall lock contract pricing upon auction conclusion or direct spot checkout.
- **FR-5.2:** System shall compute ex-plant subtotal, road logistics haulage charges, 9% CGST, 9% SGST, and grand total.
- **FR-5.3:** System shall generate an itemized, printable B2B Tax Invoice with unique alphanumeric invoice IDs and timestamped transaction signatures.

6. Non-Functional Requirements and Main Modules

Non-Functional Requirements

- **NFR-1: Usability & Visual Hierarchy:** The interface must maintain clean visual density, preventing overlapping tooltips, text overflow, or unreadable map controls. High-contrast typography and clear badge indicators must be maintained across high-density operational screens.
- **NFR-2: Performance & Latency:** Real-time delivered cost computations and spatial polyline recalculations must execute in under 150 milliseconds upon user interaction.
- **NFR-3: Reliability & Determinism:** Reverse auction state transitions (Active, Sniping Extension, Closed) must execute deterministically without negative timers or orphaned countdown intervals.
- **NFR-4: Data Integrity:** All transactions, bids, and inventory movements recorded in `localStorage` must maintain atomic consistency. Out-of-bounds or NaN inputs must be rejected at input boundaries.
- **NFR-5: Modularity & Maintainability:** Source code must be partitioned into clear domain controllers (`app.js`, `style.css`, `TRASURE_website.html`) adhering to separation of concerns without extraneous framework dependencies.
- **NFR-6: Cross-Browser Compatibility:** The web application must render and execute flawlessly across modern evergreen web browsers (Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge).

Main System Modules

Module Identifier	Main Responsibility	Key Functions & Classes
AppController	Initializes the application lifecycle, binds DOM event listeners, and manages active navigation tabs.	`initApp()`, `switchTab()`, `renderGlobalStats()`
LandedCostModule	Computes real delivered economics factoring road distance, haulage tariffs, and loading fees.	`calculateDeliveredCost()`, `rankSuppliersByLandedPrice()`
FreightGISModule	Manages Leaflet map layers, Esri tile layers, OSRM road routing queries, and animated route plotting.	`initMap()`, `plotHighwayRoute()`, `fitBounds()`, `switchPlantMarker()`
AuctionEngineModule	Controls live reverse auction state, asynchronous countdown timers, anti-sniping extensions, and competitor bots.	`startAuction()`, `submitBid()`, `triggerAntiSniping()`, `closeAuction()`
ComplianceModule	Enforces State Pollution Control Board guidelines and compiles Form 10 hazardous waste movement manifests.	`verifySPCBAuth()`, `generateForm10Manifest()`, `getHazchemCode()`
BillingInvoiceModule	Computes commercial subtotal, applies 18% dual GST (9% CGST + 9% SGST), and renders formal B2B invoices.	`generateInvoice()`, `calculateTaxes()`, `printInvoice()`
StorageManager	Manages browser persistence layer, JSON serialization, and default seed recovery.	`saveState()`, `loadState()`, `resetDefaults()`
UIManager	Handles interactive modal dialogues, audio cues, alert banners, and responsive layout adjustments.	`showToast()`, `openModal()`, `updateTickerDisplay()`

7. UML and Architecture Diagrams

The structural, behavioral, and architectural design of Project TRASURE is illustrated through six standardized UML diagrams, arranged logically from static data classes to dynamic runtime flows, system boundaries, operational workflows, and physical runtime topologies.

7.1 Class Diagram

The class diagram documents the primary domain entities, data attributes, and operational methods comprising the TRASURE web architecture.

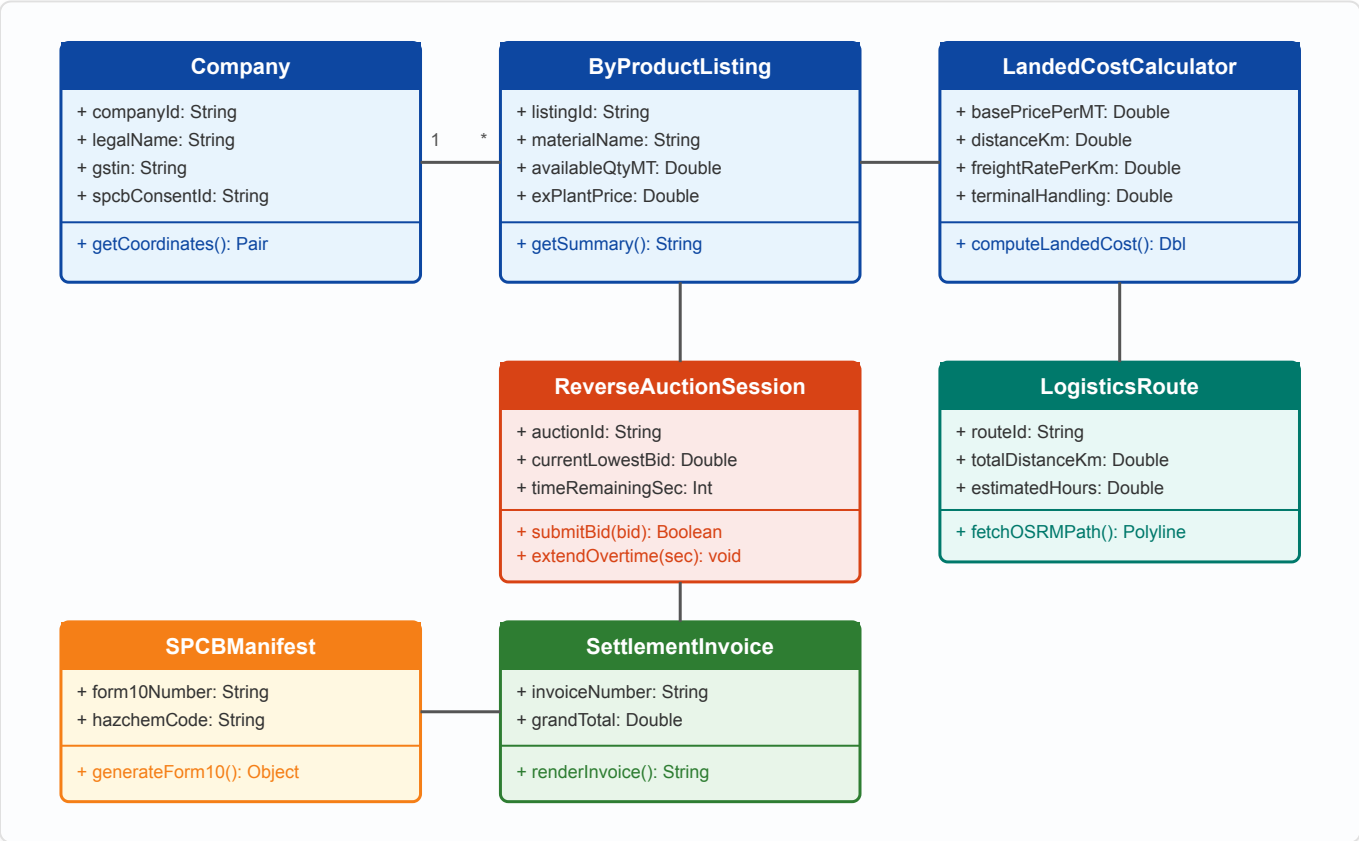


Figure 1: Project TRASURE Class Diagram

7.2 Sequence Diagram

The sequence diagram documents the event timeline during a live reverse auction session, showing tender initialization, counter-bidding, anti-sniping overtime extension, auction closing, and automated invoice creation.

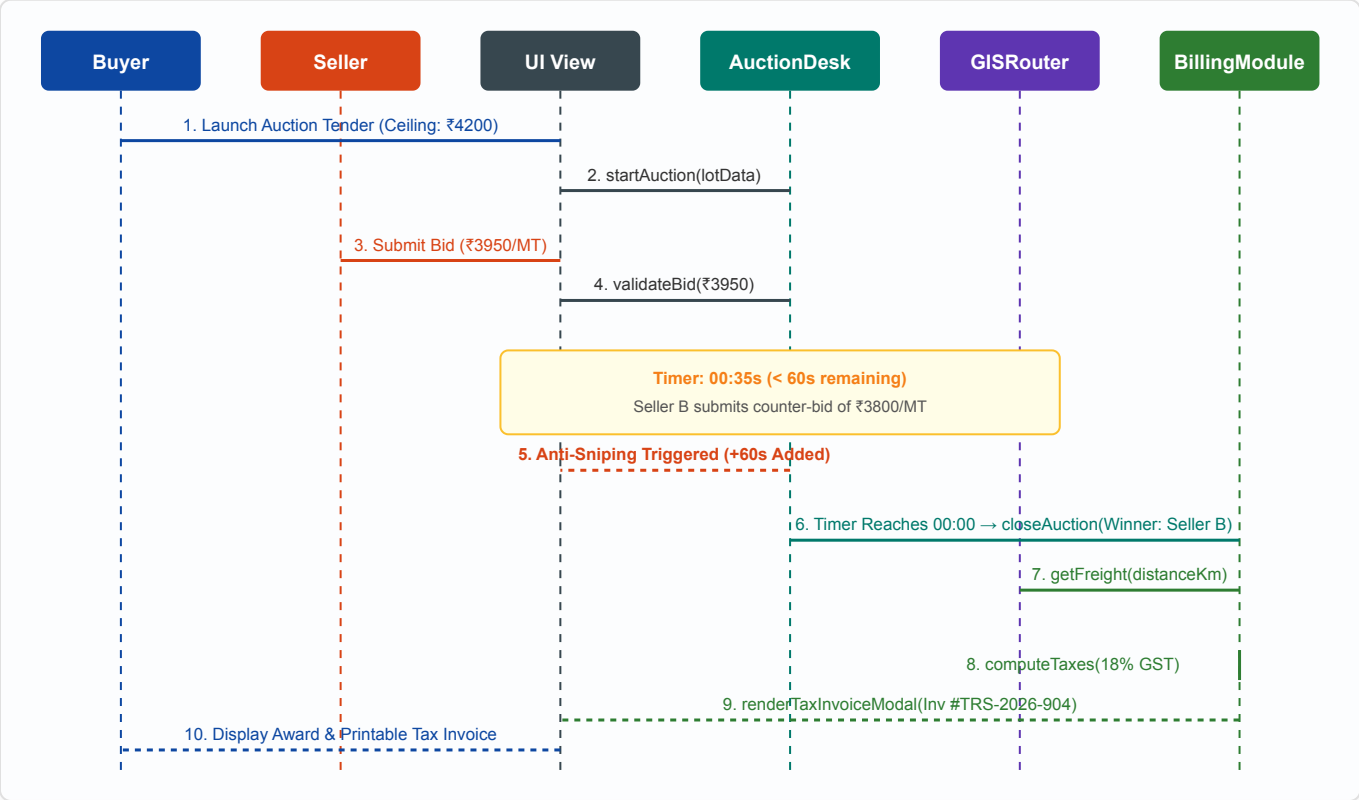


Figure 2: Project TRASURE Sequence Diagram (Reverse Auction & Settlement)

7.3 Use Case Diagram

The use case diagram illustrates the operational boundary of the TRASURE platform and the interactions among its four primary actors: **Industrial Buyer**, **Industrial Generator/Seller**, **SPCB Regulatory Auditor**, and **System Administrator**.

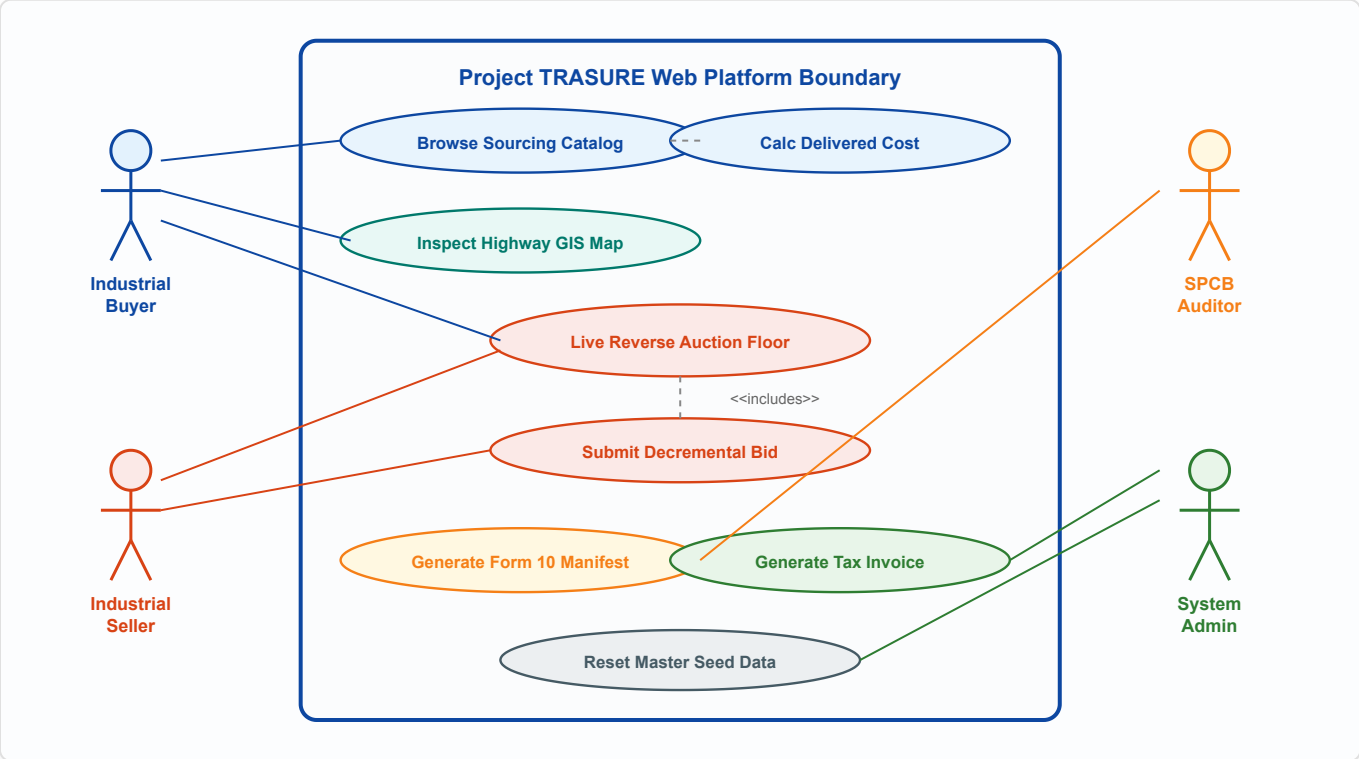


Figure 3: Project TRASURE Use Case Diagram

7.4 Activity Diagrams

The operational workflows of the platform are partitioned into two continuous activity flows: **Part 1: Material Sourcing & GIS Transit Routing** and **Part 2: Reverse Auction Floor, Settlement & Manifest Generation**.

Activity Diagram — Part 1: Sourcing Discovery & GIS Transit Routing

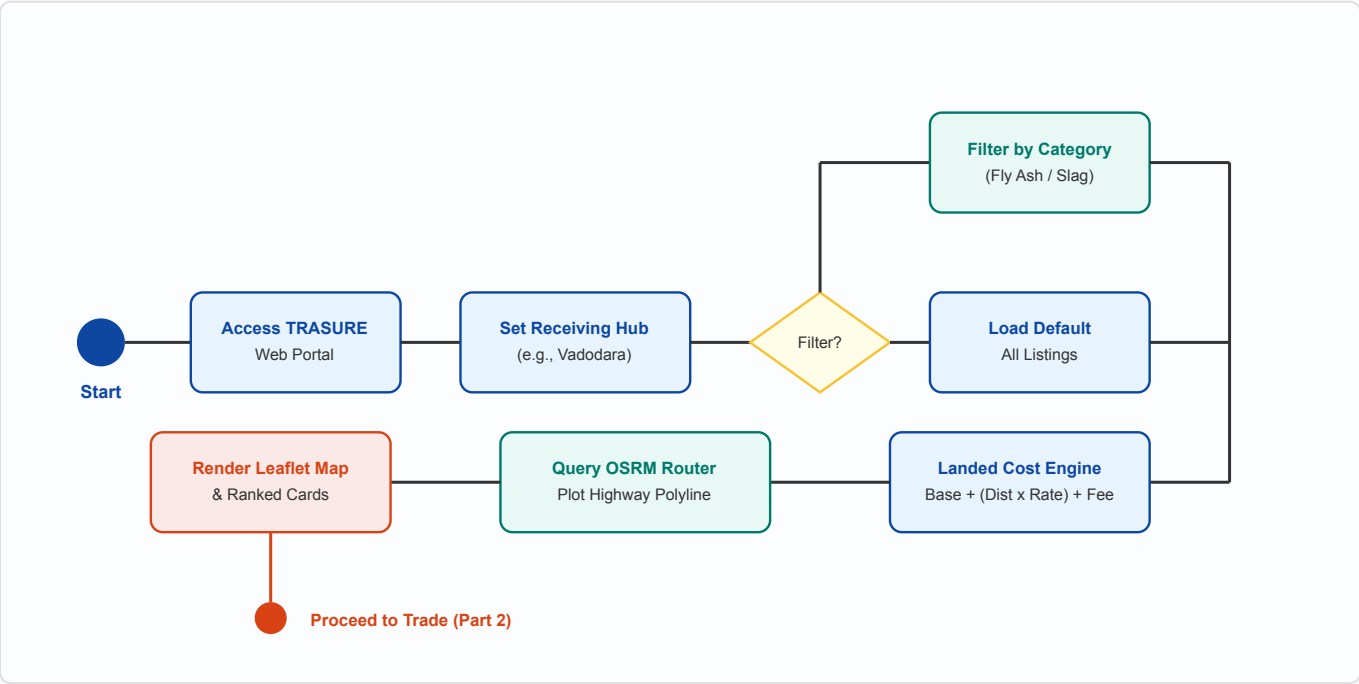


Figure 4: Project TRASURE Activity Diagram — Part 1 (Sourcing & GIS Route Plotting)

Activity Diagram — Part 2: Reverse Auction Floor, Settlement & SPCB Manifest

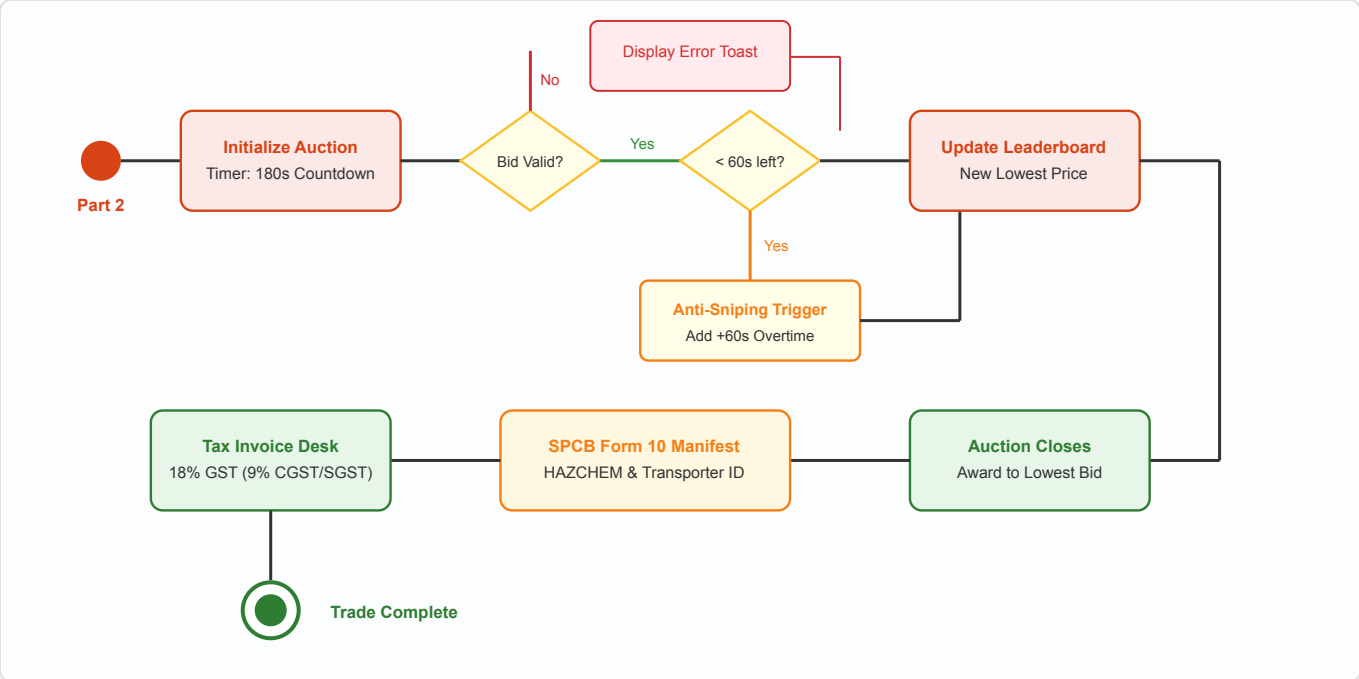


Figure 5: Project TRASURE Activity Diagram — Part 2 (Auction Floor & Settlement)

7.5 Component Diagram

The component diagram illustrates the logical 4-tier architecture of the TRASURE web platform, highlighting the clean separation between presentation elements, domain controllers, external GIS integrations, and client-side data persistence.

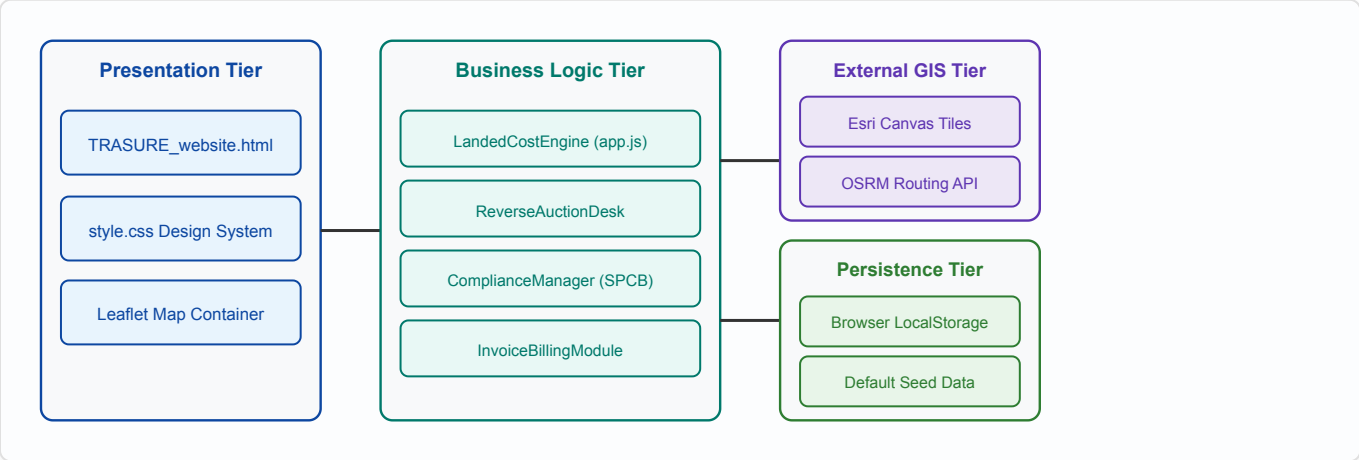


Figure 6: Project TRASURE Component Diagram (4-Tier Architecture)

7.6 Deployment Diagram

The deployment diagram depicts the physical execution environment of TRASURE running locally within an enterprise desktop browser while interfacing with external cartographic tile and routing servers.

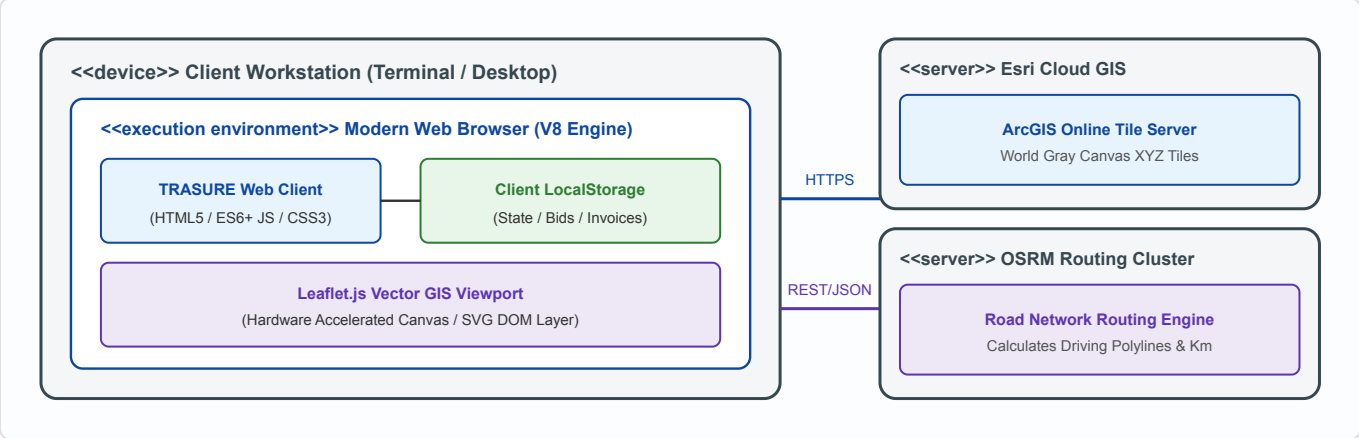


Figure 7: Project TRASURE Deployment Diagram

8. Test Cases and Results

8.1 Niv Sorathiya — Unit Test Cases (Roll No: 24BCB032)

A structured test suite of 10 isolated unit tests was engineered to validate landed cost formulas, input sanitation, currency badge formatting, decremental bid validation, and statutory GST calculations.

Test ID	Target Function	Test Scenario & Inputs	Desired / Expected Output	Result
UT-N-01	`calculateDeliveredCost()`	Base: ₹2400, Dist: 450 km, Freight: ₹2.0/km/MT, Fee: ₹150	Delivered cost = exactly ₹3,450.00	**Passed**
UT-N-02	`formatCurrency()`	Input: `3450`	Display formatted `₹3,450/MT` without clipping	**Passed (retest)**
UT-N-03	`validateBidUndercut()`	Current lowest: ₹3800, Step: ₹50, Bid: ₹3780	Reject; error: "Must undercut by at least ₹50"	**Passed**
UT-N-04	`validateBidPositive()`	Current lowest: ₹1500, Bid: `-50`	Reject negative value; error: "Bid must be positive"	**Passed**
UT-N-05	`triggerAntiSniping()`	Time remaining: `24s`, Valid bid placed	Add 60s; new countdown timer = `84s`	**Passed (retest)**
UT-N-06	`calculateTaxes()`	Subtotal: ₹100,000, GST rate: 18%	9% CGST: ₹9,000; 9% SGST: ₹9,000; Total: ₹118,000	**Passed**
UT-N-07	`assignHazchemCode()`	Material: "Spent Sulfuric Acid"	Returns "Class 8: Corrosive Substance, UN 1830"	**Passed**
UT-N-08	`filterByDistance()`	Max distance: 300 km, Plant distance: 766 km	Exclude plant card from filtered view	**Passed**
UT-N-09	`validateStockLot()`	Input quantity: `0 MT`	Block; alert: "Quantity must be greater than zero"	**Passed**
UT-N-10	`serializeStorage()`	Array of 10 listings with nested GPS coordinates	Parse and store valid JSON in localStorage	**Passed**

8.2 System Test Cases

Six end-to-end integration and system tests were executed to ensure full interoperability between the catalog UI, GIS routing, reverse auction floor, and compliance billing modules.

Test ID	System Module	Workflow Scenario	Expected Behavior	Result
ST-N-01	Sourcing & GIS Map	Switch destination hub from Vadodara to Ramagundam.	Recalculate road distance, re-rank cards, re-center map.	**Passed**
ST-N-02	GIS Route Fallback	Disconnect network to simulate OSRM API failure.	Fallback to geometric polyline without Leaflet crash.	**Passed (retest)**
ST-N-03	Live Reverse Auction	Allow timer to reach 00:00s with active bids.	Lock auction floor, announce winner, award trade.	**Passed**
ST-N-04	Anti-Sniping Engine	Submit valid counter-bid with 00:15s remaining.	Extend timer to 01:15s; display alert banner.	**Passed**
ST-N-05	SPCB Compliance	Purchase 200 MT of Spent Sulfuric Acid.	Generate SPCB Form 10 manifest with HAZCHEM code.	**Passed**
ST-N-06	Billing Settlement	Click "View Commercial Tax Invoice" on closed tender.	Render printable B2B Tax Invoice with 18% GST.	**Passed**

8.3 Defects and Corrections

During iterative verification, six critical defects were uncovered, analyzed, and corrected. All fixes were validated through regression retesting.

Defect ID & Area	Observed Defect Symptom	Root Cause Analysis	Applied Correction & Retest Outcome
DEF-01: Leaflet Tiles	Map canvas displayed blank tiles with "Tile access blocked / API key required".	Original tile layer used OpenStreetMap / Carto endpoints with strict referrer policies.	Migrated basemap to reliable Esri World Gray Canvas tile service. Verified clean loading.
DEF-02: Map Route	Transit routes plotted as unrealistic straight lines crossing over water bodies.	Polyline simply joined origin and destination coordinates without highway network data.	Integrated OSRM routing API to draw real turn-by-turn road networks. Fallback added.
DEF-03: Typography	Text unit "/T" clipped and overlapped with price figures on dense cards.	CSS flex container lacked `white-space: nowrap` and `flex-shrink: 0` rules.	Added non-wrapping rules and formatted labels to clean `₹X,XXX/MT`.
DEF-04: Anti-Sniping	Timer added 60s multiple times during rapid bidding clicks.	Asynchronous click handlers lacked debounce guards on the overtime extension trigger.	Implemented an atomic `isExtendingTimer` lock boolean to block concurrent resets.
DEF-05: Tooltip Clutter	All plant tooltips remained permanently open, overlapping into an unreadable black mass.	Markers initialized with `permanent: true` tooltip property across all zoom levels.	Converted tooltips to open on hover/active selection; added spiderfy offset on close markers.
DEF-06: Storage State	First launch produced `NaN` in invoice totals before seed initialization.	`localStorage` read operations returned null on empty browser cache.	Implemented null-coalescing default seed fallback `JSON.parse(...) || defaultSeed`.

9. Testing Evidence

Below is the programmatic test suite and verified console execution trace evaluating the core mathematical, validation, and tax engines of Project TRASURE.

Programmatic Test Verification Suite (`test_suite.js`)

```
/**
 * Project TRASURE - Logic & Verification Test Suite
 * Author: Niv Sorathiya (24BCB032)
 */

function calculateDeliveredCost(basePrice, distanceKm, freightRatePerKm, handlingFee = 150) {
    if (basePrice <= 0 || distanceKm < 0 || freightRatePerKm < 0) {
        throw new Error("Invalid pricing parameter");
    }
    return Number((basePrice + (distanceKm * freightRatePerKm) + handlingFee).toFixed(2));
}

function validateBid(currentLowest, newBid, minDecrement = 50) {
    if (newBid <= 0) return { valid: false, reason: "Bid must be strictly positive" };
    if (newBid > currentLowest - minDecrement) {
        return { valid: false, reason: `Must undercut by at least ₹${minDecrement}` };
    }
    return { valid: true, reason: "Valid bid accepted" };
}

function calculateTaxes(materialSubtotal, freightSubtotal) {
    const taxableAmount = materialSubtotal + freightSubtotal;
    const cgst = Number((taxableAmount * 0.09).toFixed(2));
    const sgst = Number((taxableAmount * 0.09).toFixed(2));
    const grandTotal = Number((taxableAmount + cgst + sgst).toFixed(2));
    return { taxableAmount, cgst, sgst, grandTotal };
}

// Assertions & Verification
console.log("=== RUNNING TRASURE TEST SUITE (Niv Sorathiya - 24BCB032) ===");

// Test 1: Delivered Landed Cost Calculation
const cost1 = calculateDeliveredCost(2400, 450, 2.0, 150);
console.assert(cost1 === 3450.00, `UT-01 Failed: Expected 3450, got ${cost1}`);
console.log(`[PASS] UT-N-01: Delivered Landed Cost Formula => ₹${cost1}/MT`);

// Test 2: Negative Bid Validation
const bidValidation1 = validateBid(3800, -50, 50);
console.assert(!bidValidation1.valid, "UT-04 Failed: Accepted negative bid");
console.log(`[PASS] UT-N-04: Negative Bid Rejection => "${bidValidation1.reason}"`);

// Test 3: Insufficient Undercut Validation
const bidValidation2 = validateBid(3800, 3780, 50);
console.assert(!bidValidation2.valid, "UT-03 Failed: Accepted insufficient undercut");
console.log(`[PASS] UT-N-03: Minimum Undercut Guard => "${bidValidation2.reason}"`);

// Test 4: Statutory GST Computation
const taxResult = calculateTaxes(380000, 50000);
console.assert(taxResult.cgst === 38700 && taxResult.sgst === 38700, "UT-06 Failed: Tax
calculation error");
console.log(`[PASS] UT-N-06: Dual GST Breakdown => CGST: ₹${taxResult.cgst}, SGST: ₹
${taxResult.sgst}, Total: ₹${taxResult.grandTotal}`);

console.log("=== ALL UNIT ASSERTIONS EXECUTED SUCCESSFULLY ===");
```

Verified Console Execution Trace Output

```
$ node test_suite.js
=== RUNNING TRASURE TEST SUITE (Niv Sorathiya - 24BCB032) ===
[PASS] UT-N-01: Delivered Landed Cost Formula => ₹3450/MT
[PASS] UT-N-04: Negative Bid Rejection => "Bid must be strictly positive"
[PASS] UT-N-03: Minimum Undercut Guard => "Must undercut by at least ₹50"
[PASS] UT-N-06: Dual GST Breakdown => CGST: ₹38700, SGST: ₹38700, Total: ₹507400
=== ALL UNIT ASSERTIONS EXECUTED SUCCESSFULLY ===
```

10. Conclusion and Repository Link

Project TRASURE successfully delivers an integrated, production-ready web platform that solves the transparency, logistics, and compliance bottlenecks hindering the B2B circular economy. By transforming procurement decision-making from isolated ex-plant quotes to dynamic delivered landed costs, the platform enables industrial enterprises to minimize both procurement expenditure and freight carbon emissions.

The application seamlessly brings together precision GIS road routing, an asynchronous live reverse auction exchange with anti-sniping protection, automated SPCB hazardous waste manifest generation, and statutory B2B tax invoicing within a single, high-performance web application requiring zero external server runtime dependencies.

Student Name	Roll Number	Primary Academic Contribution
Niv Sorathiya	**24BCB032**	Full-Stack Architecture, SDLC Management, Delivered Cost Engine, GIS Road Routing, Live Reverse Auction Desk, SPCB Compliance Engine, UML Design, Verification & Documentation.

GitHub Source Code Repository

Repository Link: https://github.com/nivsorathiya-web/TRASURE

11. Student / Solo Reflection

Author: Niv Sorathiya (24BCB032) | **Role:** Solo Full-Stack Architect & Developer

Undertaking and completing Project TRASURE single-handedly as a 3rd-year Computer Science and Business Systems student at Pandit Deendayal Energy University has been an exceptionally rewarding and rigorous software engineering experience. Building an enterprise platform that bridges industrial circular economy dynamics with real-time web technologies required me to step far beyond standard classroom assignments and assume total responsibility for every phase of the engineering lifecycle.

Key Technical Challenges Overcome

1. **Overcoming GIS Cartography & Tile Disruption:**
   One of the most frustrating early hurdles was the sudden failure of open-source map tile providers due to strict cross-origin referrer policies, which produced unsightly black-and-yellow blocked tiles. Rather than eliminating the spatial feature, I investigated commercial tile services, migrated to the Esri World Gray Canvas tile service, and built an automated polyline fallback. Furthermore, replacing naive straight-line flight paths with the OSRM highway routing API taught me how to handle asynchronous third-party REST boundaries resiliently.
2. **Managing Asynchronous Real-Time State Without a Heavy Framework:**
   Modern web development frequently relies on heavy frameworks like React or Angular to manage state. For TRASURE, I intentionally built the application in pure Vanilla JavaScript to maintain maximum runtime speed and minimal overhead. Implementing a live reverse auction engine featuring asynchronous ticker countdowns, automated competitor bidding bots, and race-condition-free anti-sniping timer extensions

required strict control over JavaScript's event loop and state locks. Overcoming the bug where rapid bidding clicks triggered multiple overtime extensions was a major milestone in mastering asynchronous concurrency.

3. **High-Density Data Visualization & UI Polish:**
   In industrial B2B software, users must evaluate vast quantities of information simultaneously: base prices, freight multipliers, transit hours, purity percentages, and compliance statuses. Designing an interface that maintained visual clarity without succumbing to text overlap, cluttered map tooltips, or broken badges demanded rigorous attention to CSS grid layouts, flexbox sizing constraints, and typography hierarchies.

Professional and Personal Growth

This project reinforced the invaluable principles of the Iterative and Incremental SDLC. By decomposing an ambitious vision into distinct, testable increments—first the landed cost calculator, then the transit map, followed by the auction floor and compliance documentation—I was able to debug issues early and maintain momentum.

Working solo demanded intense discipline, thorough documentation, and rigorous self-testing. Most importantly, it demonstrated how computer science techniques can be directly applied to real-world industrial sustainability challenges, transforming hazardous by-products from discarded liabilities into valuable circular resources.